

Haoyang Hong

Ph.D. Student in Artificial Intelligence, Oregon State University

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Research Interests

Reinforcement learning theory, bandit theory, online learning, robustness, statistical inference under adaptive data collection.

Education

Oregon State University

Sep. 2024 – Present

Ph.D. in Artificial Intelligence

Advisor: Huazheng Wang

University of Science and Technology of China

Sep. 2020 – Jul. 2024

B.S. in Statistics, School of the Gifted Young

Publications

Online KL-Regularized Reinforcement Learning with Function Approximation under Misspecification

Haoyang Hong*, Zichen Wang*, Quanquan Gu, Huazheng Wang

RLC 2026

When Can You Poison Rewards? A Tight Characterization of Reward Poisoning in Linear MDPs

Jose Efraim Aguilar Escamilla*, Haoyang Hong*, Jiawei Li*, Haoyu Zhao, Xuezhou Zhang, Sanghyun Hong, Huazheng Wang

ICML 2026

Design-Based Bandits under Network Interference: A Trade-off between Regret and Inference

Zichen Wang*, Haoyang Hong*, Chuanhao Li, Haoxuan Li, Zhiheng Zhang, Huazheng Wang

NeurIPS 2025

*Equal contribution.

Selected Research Directions

Robust and Regularized Reinforcement Learning

- Study how regularization, reference policies, and model misspecification reshape regret guarantees and policy benchmarks in online reinforcement learning.
- Analyze instance-level vulnerability and robustness of reinforcement learning systems under structured reward perturbations.

Bandit Learning with Inference Constraints

- Develop online learning algorithms that balance regret minimization with valid statistical inference under adaptive data collection.
- Study bandit settings with interference, preference feedback, and forced exploration, with an emphasis on clean non-asymptotic guarantees.

Theory-Guided Evaluation and Reproducible Algorithms

- Build simulation and evaluation pipelines for theoretical learning algorithms, focusing on empirical verification of regret, robustness, and inference trade-offs.

Research and Technical Skills

Theory: regret analysis, high-probability analysis, online learning, bandit algorithms, reinforcement learning theory.

Implementation: Python-based algorithm prototyping, simulation design, empirical evaluation, and reproducible research pipelines.